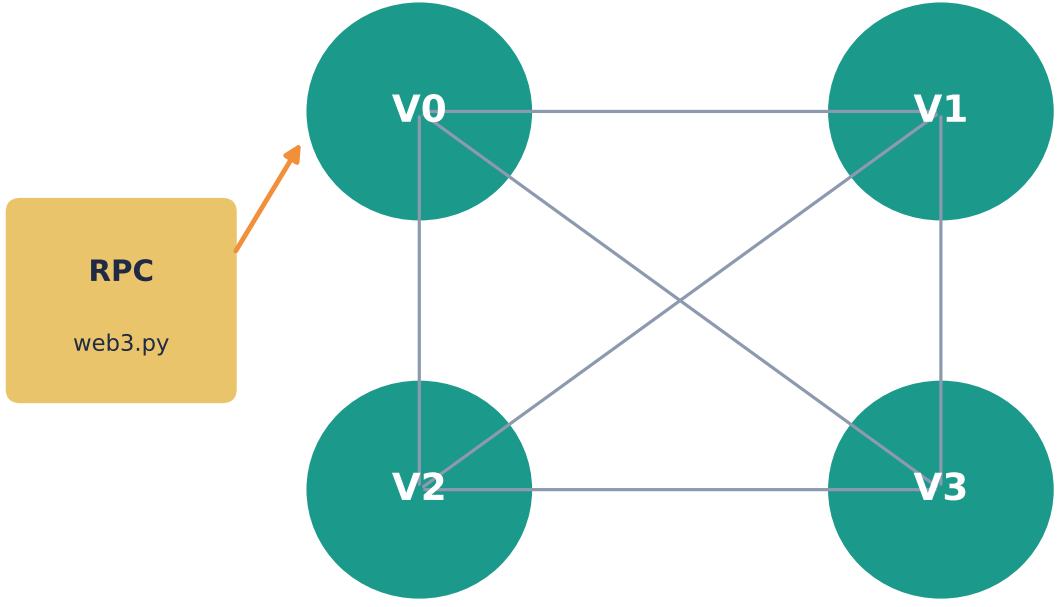
Real multi-node blockchain: 4 Hyperledger Besu validators, QBFT consensus



Fault tolerance (n = 4, f = 1, quorum 2f + 1 = 3)

all 4 up -> blocks every second, same head on every node
1 validator killed -> federation still completes
2 validators killed -> no quorum: chain halts (never forks)

Measured on this machine

4 hospitals: 23 transactions, 14-19 s per round
(1 s blocks, ~13 sequential phases)
all 66 tests pass, incl. the kill-a-validator test